MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, DIFFERENTIAL LINE, DIGITAL, KEY CODE KDM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic differential line driver module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN. ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.426 failures per million operating hours for class I or 4.021 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/55-1 for class I and M28787/55-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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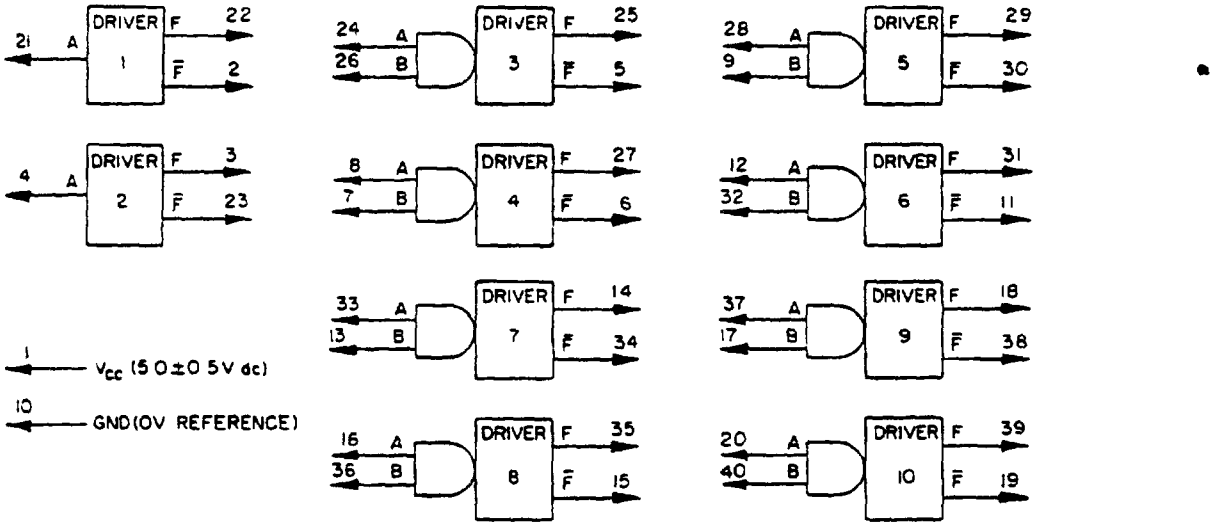


FIGURE 1. KDM module function diagram, differential line driver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A20	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	10	800	mV
B1,B2	Input breakdown current	IV	S	0	225	0	300	0	300	0	300	0	300	µA
B3-B10	Input breakdown current	IV	S	0	75	0	100	0	100	0	100	0	100	µA
B11-B18	Input breakdown current	IV	S	0	150	0	200	0	200	0	200	0	200	µA
C	Isolation													
	Initial	V	S	10	-	10	-	10	-	10	-	10	-	MΩ
	End-of-life	V	S	1	-	1	-	1	-	1	-	1	-	MΩ
D1,D2	Input capacitance	VI	Q	2	20	2	20	2	20	2	20	2	20	pF
D3-D18	Input capacitance	VI	Q	2	15	2	15	2	15	2	15	2	15	pF
E1-E18	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	0.4	-1.5	-0.4	-1.5	V
F1-F20	Output clamp diode voltage	VIII	Q	0.3	1.5	0.3	1.5	0.3	1.5	0.3	1.5	0.3	1.5	V
G1-G28	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H28	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J18	Input current, low level	XI	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
K1,K2	Input current, high level	XII	X	0	90	0	120	0	120	0	120	0	120	µA
K3-K10	Input current, high level	XII	X	0	30	0	40	0	40	0	40	0	40	µA
K11-K18	Input current, high level	XII	X	0	60	0	80	0	80	0	80	0	80	µA
L1-L20	Output short circuit current	XIII	X	40	120	40	120	40	120	40	120	30	120	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1	Power supply current, low level	XIV	S	1	244	1	244	1	244	1	244	1	244	mA
M2	Power supply current, high level	XIV	S	1	244	1	244	1	244	1	244	1	244	mA
N1-N20	Transition time (t_{TLH})	XV	S	4	45	4	50	4	50	4	50	4	50	ns
P1-P20	Transition time (t_{THL})	XV	S	2	20	2	25	2	25	2	25	2	25	ns
R1-R20	Propagation delay time (t_{PLH})	XV	S	5	30	5	45	5	45	5	45	5	45	ns
S1-S20	Propagation delay time (t_{PHL})	XV	S	5	30	5	35	5	35	5	35	5	35	ns
T1-T20	Output crosstalk, low level	XVI	Q	0	1	0	1	0	1	0	1	0	1	V
U1-U20	Output crosstalk, high level	XVII	Q	0	750	0	750	0	750	0	750	0	750	mV
V	Power supply transient amplitude	XVIII	Q	0.0	1.3	0.0	1.3	0.0	1.3	0.0	1.3	0.0	1 3	V
W	Power supply transient rise time	XIX	S	4.8	8.0	4.8	8.0	4.8	8.0	4.8	8.0	4.8	8 0	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through S and W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28 The module shall be operated as shown on figure 2. Following the life test and still at the maximum class temperature, electrical tests E through S and W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/55 supersedes Department of the Navy, Navy Ordnance Systems Command Drawing No. 10001-2658772. Part number M28787/55-1 supersedes part numbers 2658772 and M28787/55.

6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Supply voltage	5.0 V dc ±10%	5.0 V dc ±10%	
Maximum power dissipation (Outputs not loaded)	1.5	1.5	W
Input threshold voltage			
Low level	0.8 maximum	0.8 maximum	V
High level	2 minimum	2 minimum	V
Input current			
Low level (V_{IL} = 0 V)	1.73 maximum	1.73 maximum	mA
High level (V_{IH} = 2.4 V)			
1A, 2A	120 maximum	120 maximum	µA
3A-10A	40 maximum	40 maximum	µA
3B-10B	80 maximum	80 maximum	µA
Output voltage			
Low level	0.0 minimum, 0.4 maximum	0.0 minimum, 0.4 maximum	V
High level	2.4 minimum, 4.5 maximum,	2.4 minimum, 4.5 maximum	V
Output drive current			
Low level	40 maximum	40 maximum	mA
High level	10 maximum	10 maximum	mA
Propagation delay time			
t_{PLH}	5 minimum, 45 maximum	5 minimum, 45 maximum	ns
t_{PHL}	5 minimum, 35 maximum	5 minimum, 35 maximum	ns

* 6.6 Module function. This module contains 10 independent differential line drivers. Eight line drivers that have two inputs brought out to the module pins to provide the two-input NAND/AND function. Two line driver circuits have single inputs. For line driving (double-ended interface applications, this module requires the use of the differential line receiver module key code KDN (MIL-M-28787/53). Each driver circuit can drive one receiver circuit. Interconnection may be made such that the receiver output is either inverting or noninverting with respect to the driver input. This module may also be used as a high fanout logic driver with complementary outputs, or to provide static output levels. For static usage, each input is tied to ground. The static operation may be used to provide active pullup for unused inputs.

6.6.1 Input to output relationship. Each differential line driver has the input to output relationship of table II for positive logic.

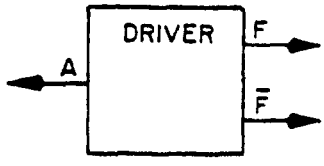
6.6.2 Parallel outputs. Paralleling of outputs is not permissible.

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TABLE II. Input to output relationship.

CIRCUITS 1-2

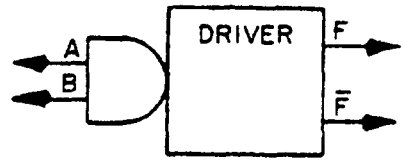
$F = A$
$\overline{F} = \overline{A}$



A	F	$\overline{F}$
H	H	L
L	L	H

CIRCUITS 3-10

$F = AB$
$\overline{F} = \overline{AB}$



A	B	F	$\overline{F}$
H	H	H	L
H	L	L	H
L	H	L	H
L	L	L	H

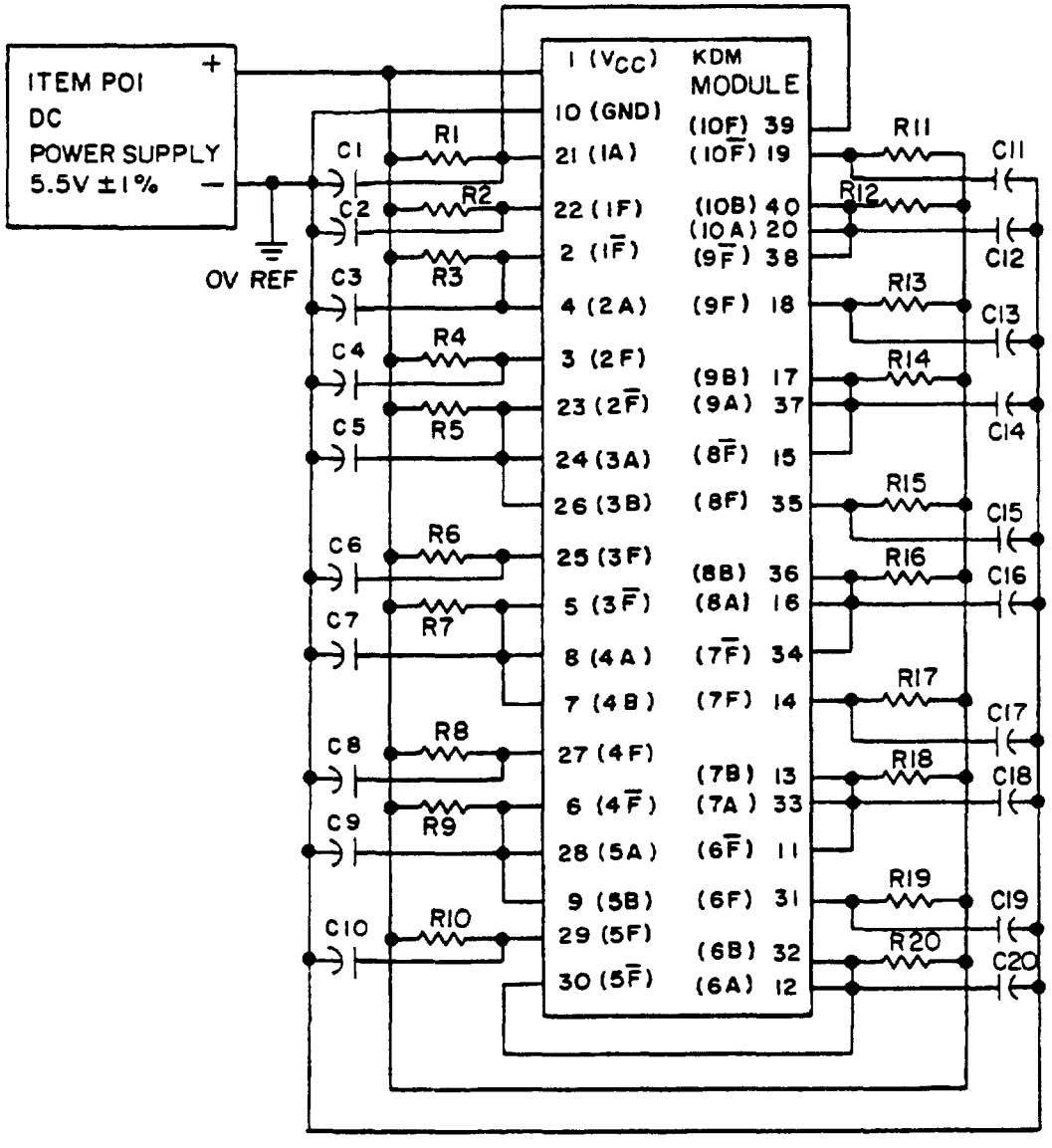
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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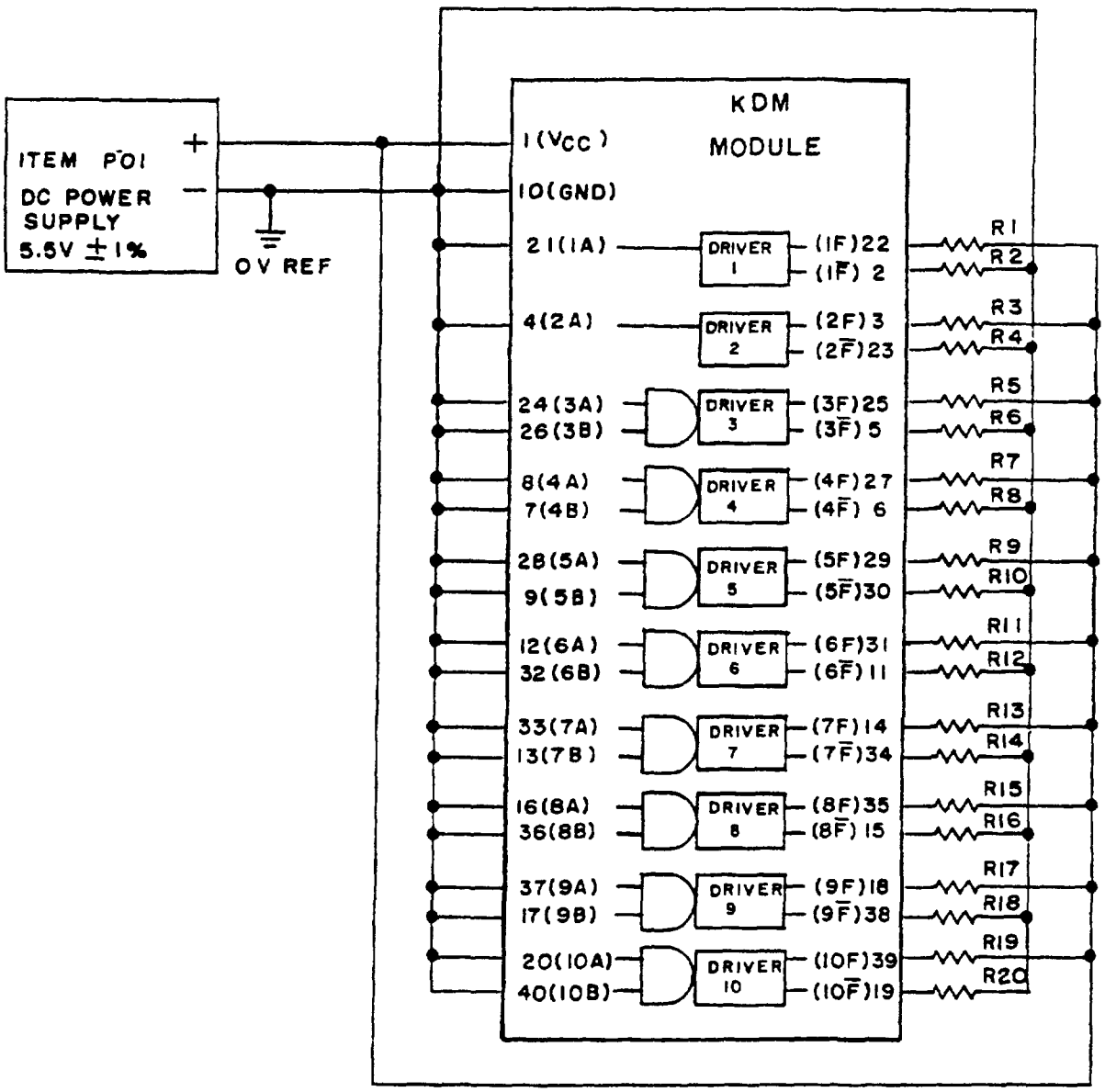
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NOTES:
1. R1-R20 = 133Ω ±1%, 1/2 W.
2. C1-C20 = 100 pF ±20%.

* FIGURE 2. Operating temperature and life test circuit.

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NOTES
1. R1 = R3 = R5 = R7 = R9 = R11 = R13 = R15 = R17 = R19 = 133Ω ±1%, 1/2 W.
2. R2 = R4 = R6 = R8 = R10 = R12 = R14 = R16 = R18 = R20 = 249Ω ±1%, 1/4 W.

* FIGURE 3. Component temperature test circuit.

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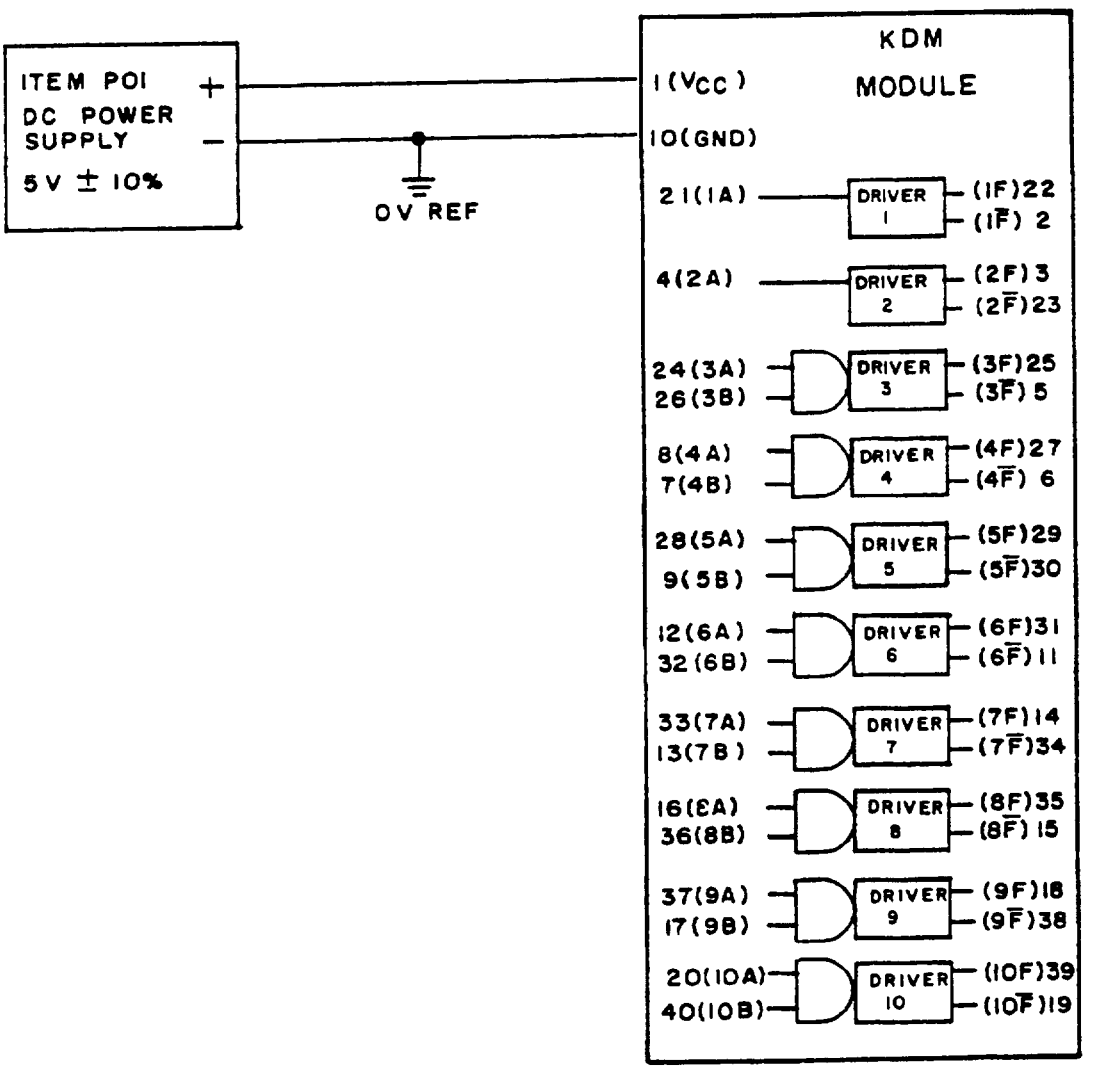


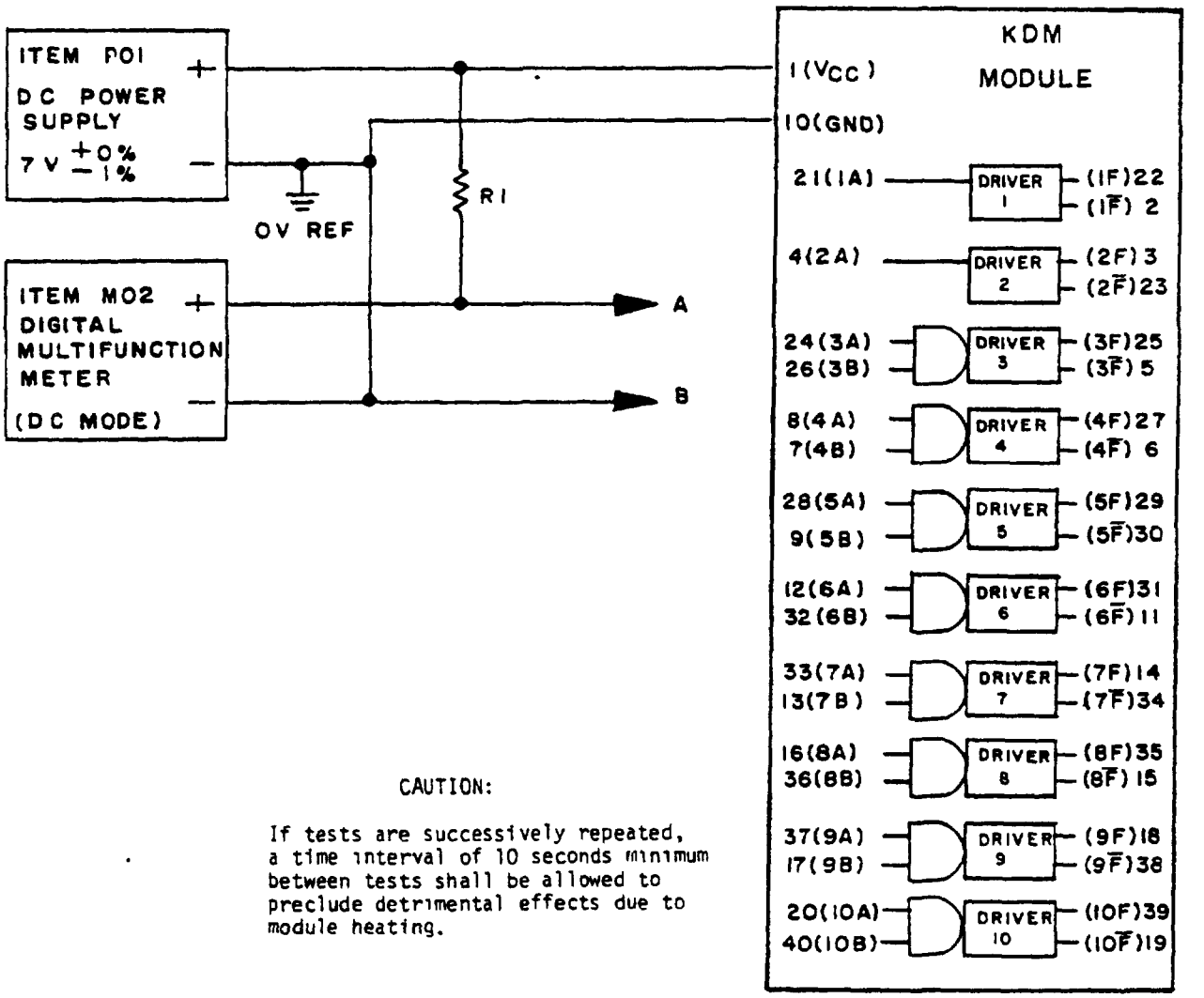
FIGURE 4. Vibration test circuit.

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* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	$1\overline{F}$ V_{OUT} (low loaded)	2	Open	10	800	mV	5
A2	$2\overline{F}$ "	23	"	"	"	"	"
A3	$3\overline{F}$ "	5	"	"	"	"	"
A4	$4\overline{F}$ "	6	"	"	"	"	"
A5	$5\overline{F}$ "	30	"	"	"	"	"
A6	$6\overline{F}$ "	11	"	"	"	"	"
A7	$7\overline{F}$ "	34	"	"	"	"	"
A8	$8\overline{F}$ "	15	"	"	"	"	"
A9	$9\overline{F}$ "	38	"	"	"	"	"
A10	$10\overline{F}$ "	19	"	"	"	"	"
A11	1F "	22	21	"	"	"	"
A12	2F "	3	4	"	"	"	"
A13	3F "	25	24	"	"	"	"
A14	4F "	27	8	"	"	"	"
A15	5F "	29	28	"	"	"	"
A16	6F "	31	12	"	"	"	"
A17	7F "	14	33	"	"	"	"
A18	8F "	35	16	"	"	"	"
A19	9F "	18	37	"	"	"	"
[illegible]	10F "	39	20	"	"	"	"

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CAUTION:

If tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

NOTE: R1 = 133Ω ±1%, 1/4 W.

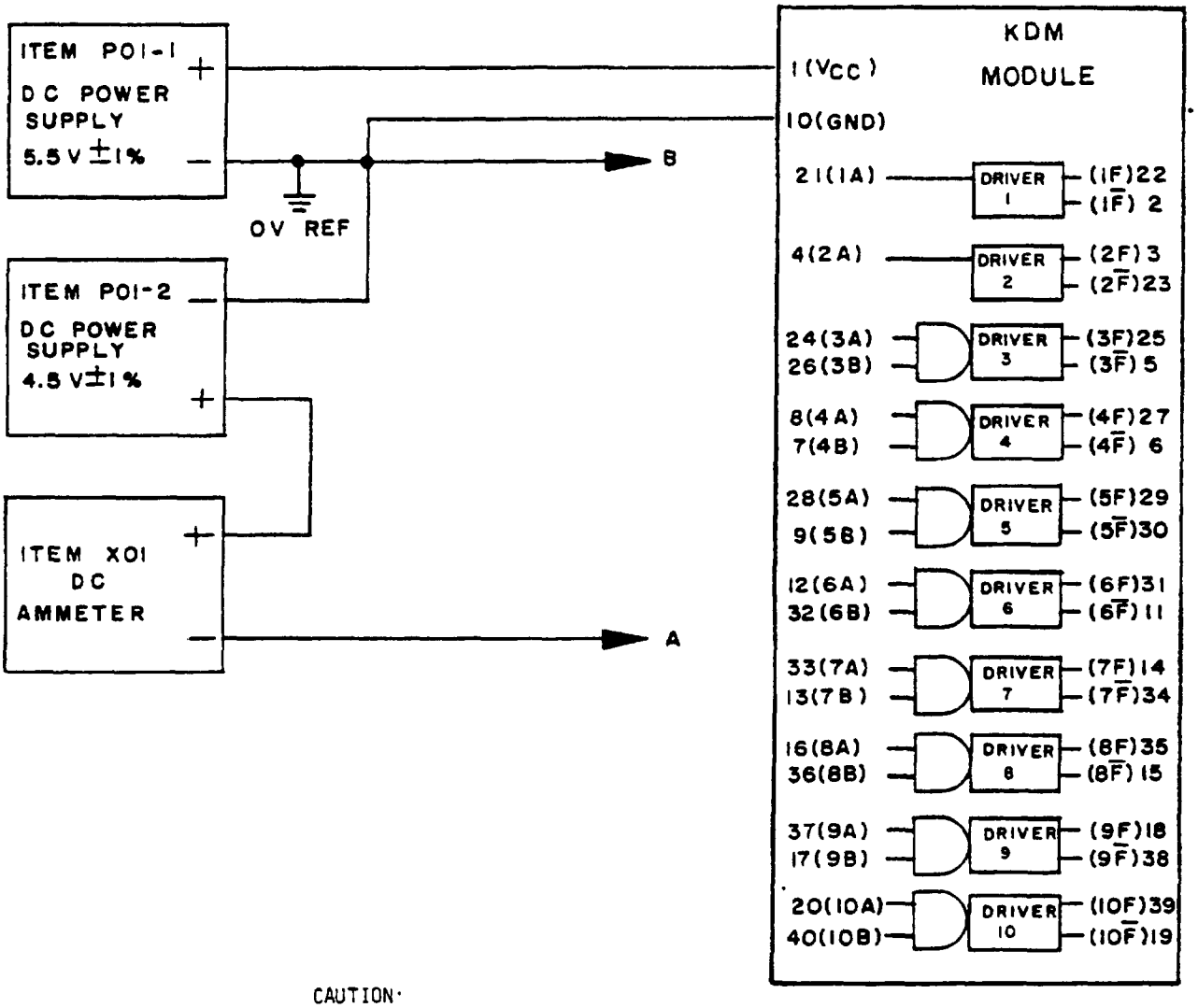
* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1A	Input breakdown current	21	Open	0	225	µA	6
B2	2A	"	4	Open	"	225	"	"
B3	3A	"	24	26	"	75	"	"
B4	4A	"	8	7	"	"	"	"
B5	5A	"	28	9	"	"	"	"
B6	6A	"	12	32	"	"	"	"
B7	7A	"	33	13	"	"	"	"
B8	8A	"	16	36	"	"	"	"
B9	9A	"	37	17	"	"	"	"
B10	10A	"	20	40	"	"	"	"
B11	3B	"	26	24	"	150	"	"
B12	4B	"	7	8	"	"	"	"
B13	5B	"	9	28	"	"	"	"
B14	6B	"	32	12	"	"	"	"
B15	7B	"	13	33	"	"	"	"
B16	8B	"	36	16	"	"	"	"
B17	9B	"	17	37	"	"	"	"
B18	10B	"	40	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground. If tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

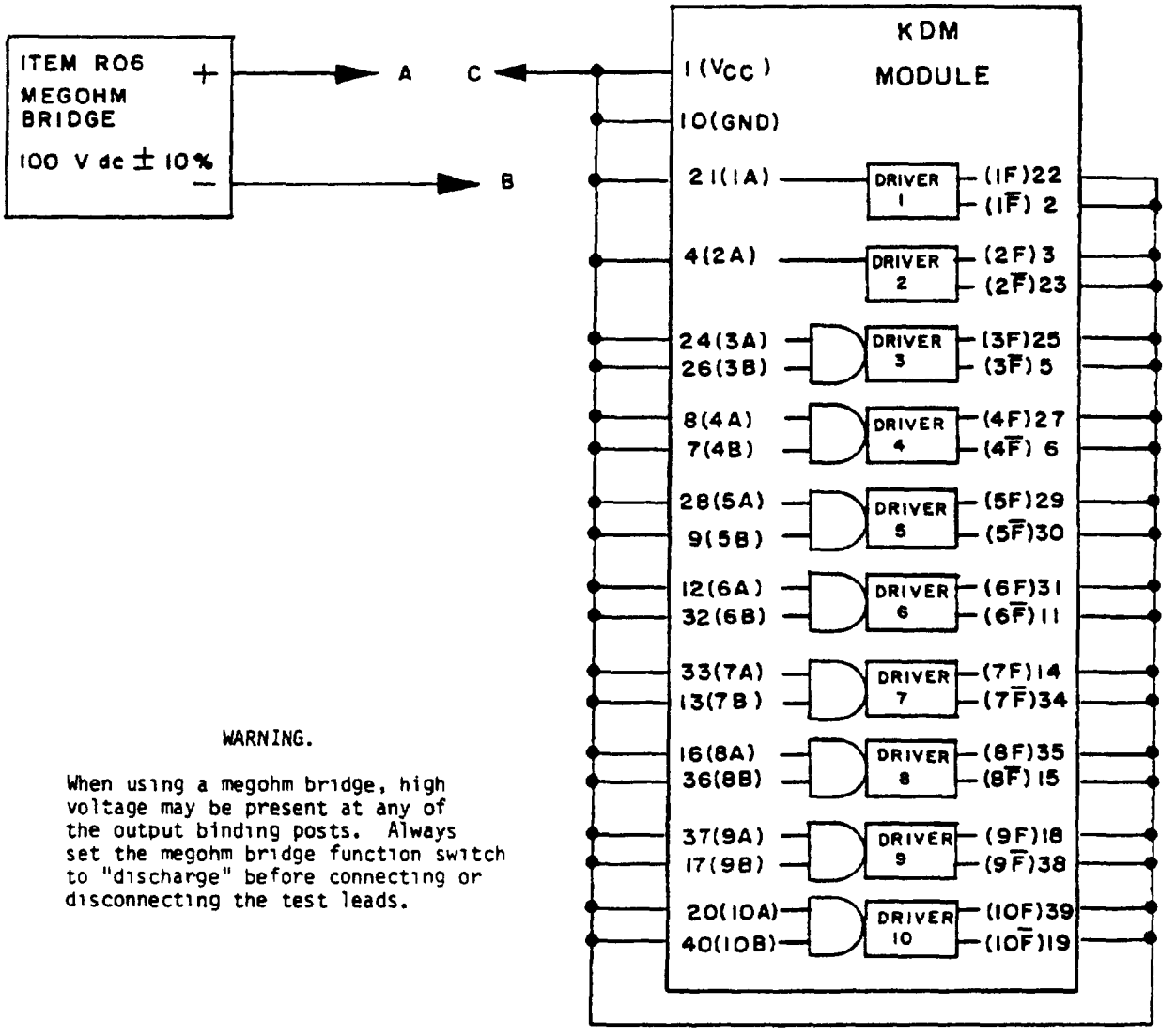
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	C	Frame	10	∞	MΩ	7

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WARNING.

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

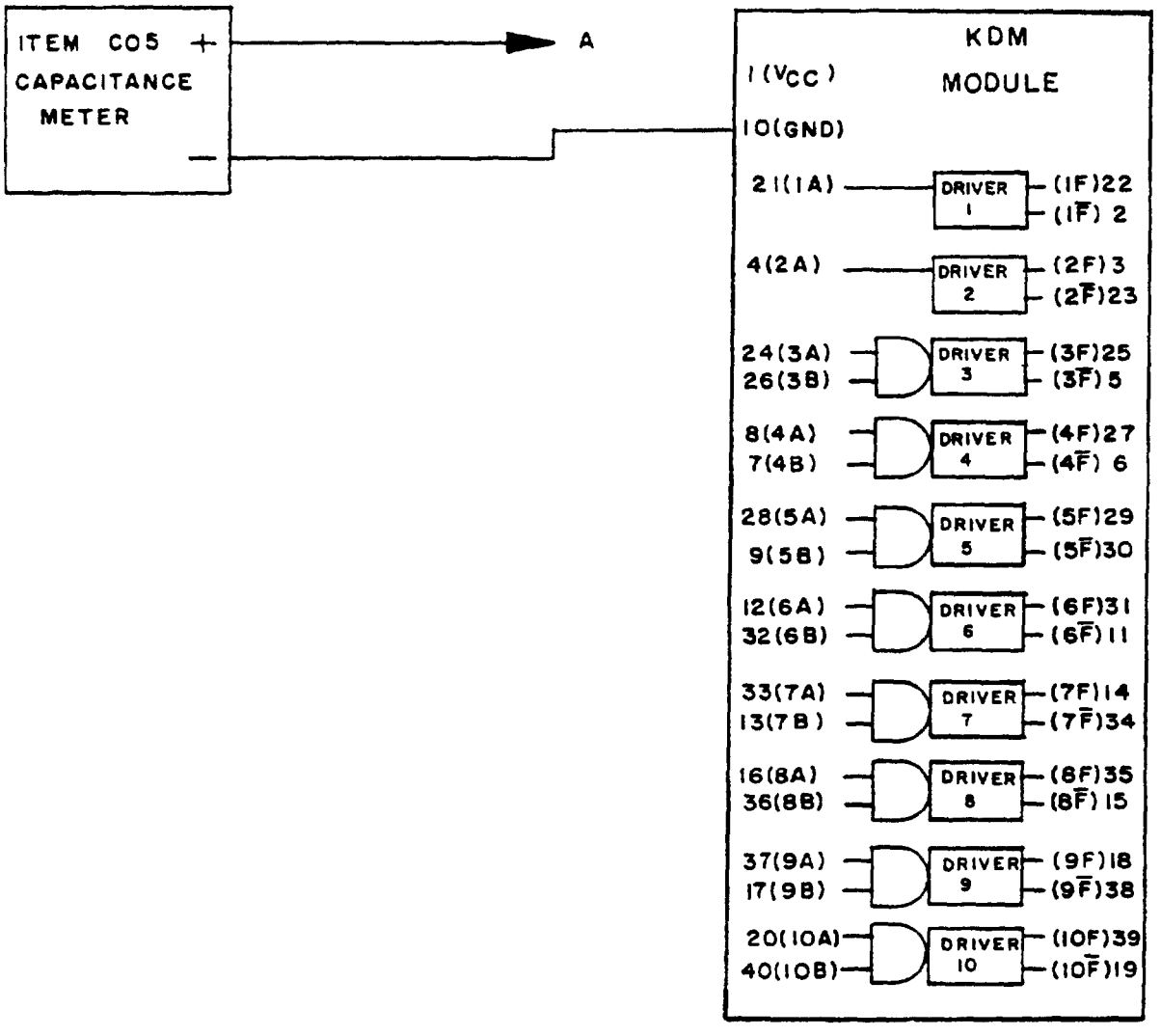
FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Input capacitance test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1A	Input capacitance	21	2	20	pF	8
D2	2A	"	4	"	20	"	"
D3	3A	"	24	"	15	"	"
D4	3B	"	26	"	"	"	"
D5	4A	"	8	"	"	"	"
D6	4B	"	7	"	"	"	"
D7	5A	"	28	"	"	"	"
D8	5B	"	9	"	"	"	"
D9	6A	"	12	"	"	"	"
D10	6B	"	32	"	"	"	"
D11	7A	"	33	"	"	"	"
D12	7B	"	13	"	"	"	"
D13	8A	"	16	"	"	"	"
D14	8B	"	36	"	"	"	"
D15	9A	"	37	"	"	"	"
D16	9B	"	17	"	"	"	"
D17	10A	"	20	"	"	"	"
D18	10B	"	40	"	"	"	"

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NOTES:
1. Null out the effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

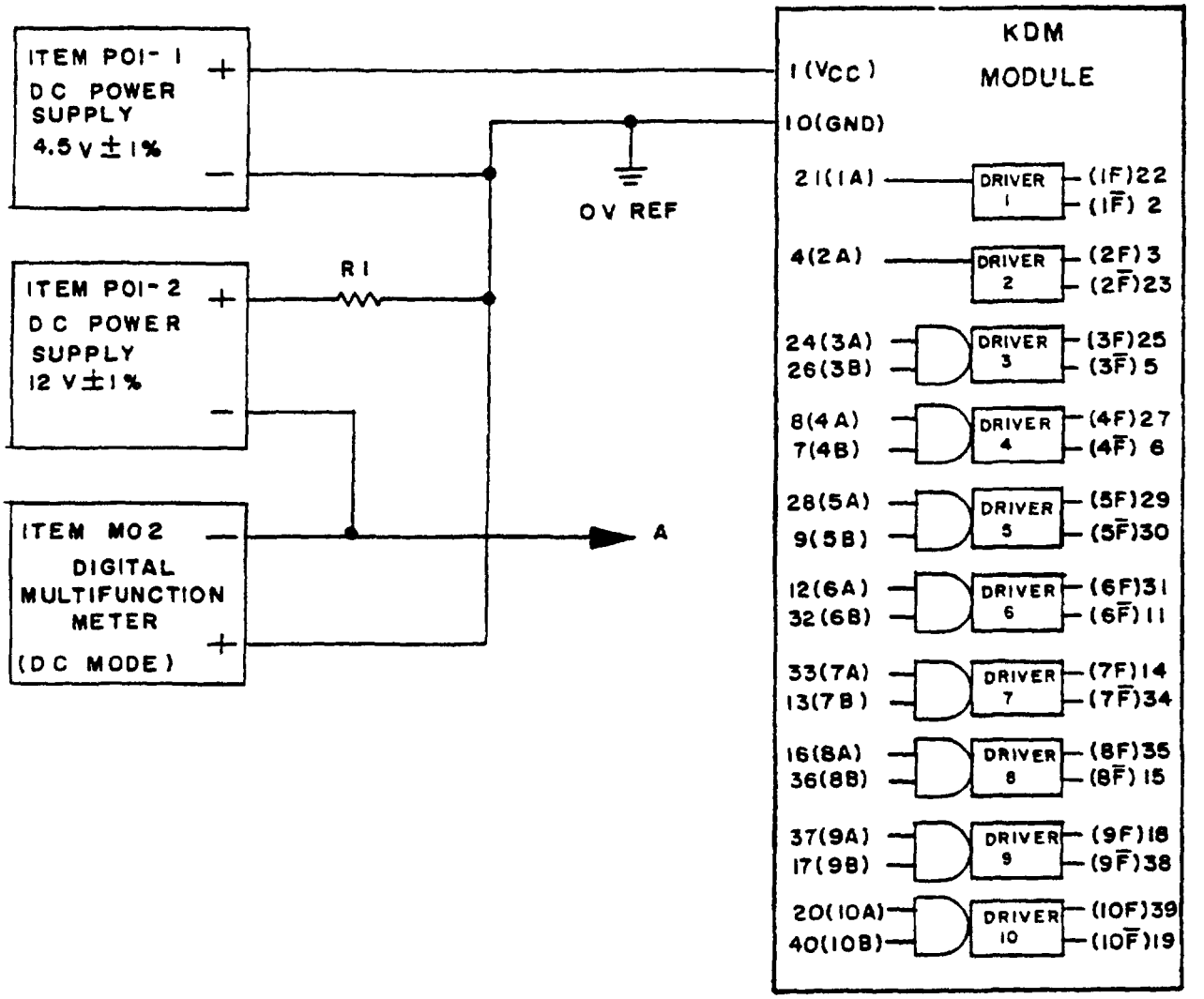
FIGURE 8. Input capacitance test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E1	1A	Input clamp diode voltage	21	-0.4	-1.5	V	9
E2	2A	"	4	"	"	"	"
E3	3A	"	24	"	"	"	"
E4	3B	"	26	"	"	"	"
E5	4A	"	8	"	"	"	"
E6	4B	"	7	"	"	"	"
E7	5A	"	28	"	"	"	"
E8	5B	"	9	"	"	"	"
E9	6A	"	12	"	"	"	"
E10	6B	"	32	"	"	"	"
E11	7A	"	33	"	"	"	"
E12	7B	"	13	"	"	"	"
E13	8A	"	16	"	"	"	"
E14	8B	"	36	"	"	"	"
E15	9A	"	37	"	"	"	"
E16	9B	"	17	"	"	"	"
E17	10A	"	20	"	"	"	"
E18	10B	"	40	"	"	"	"

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NOTE: R1 = 887Ω ±1%, 1/4 W.

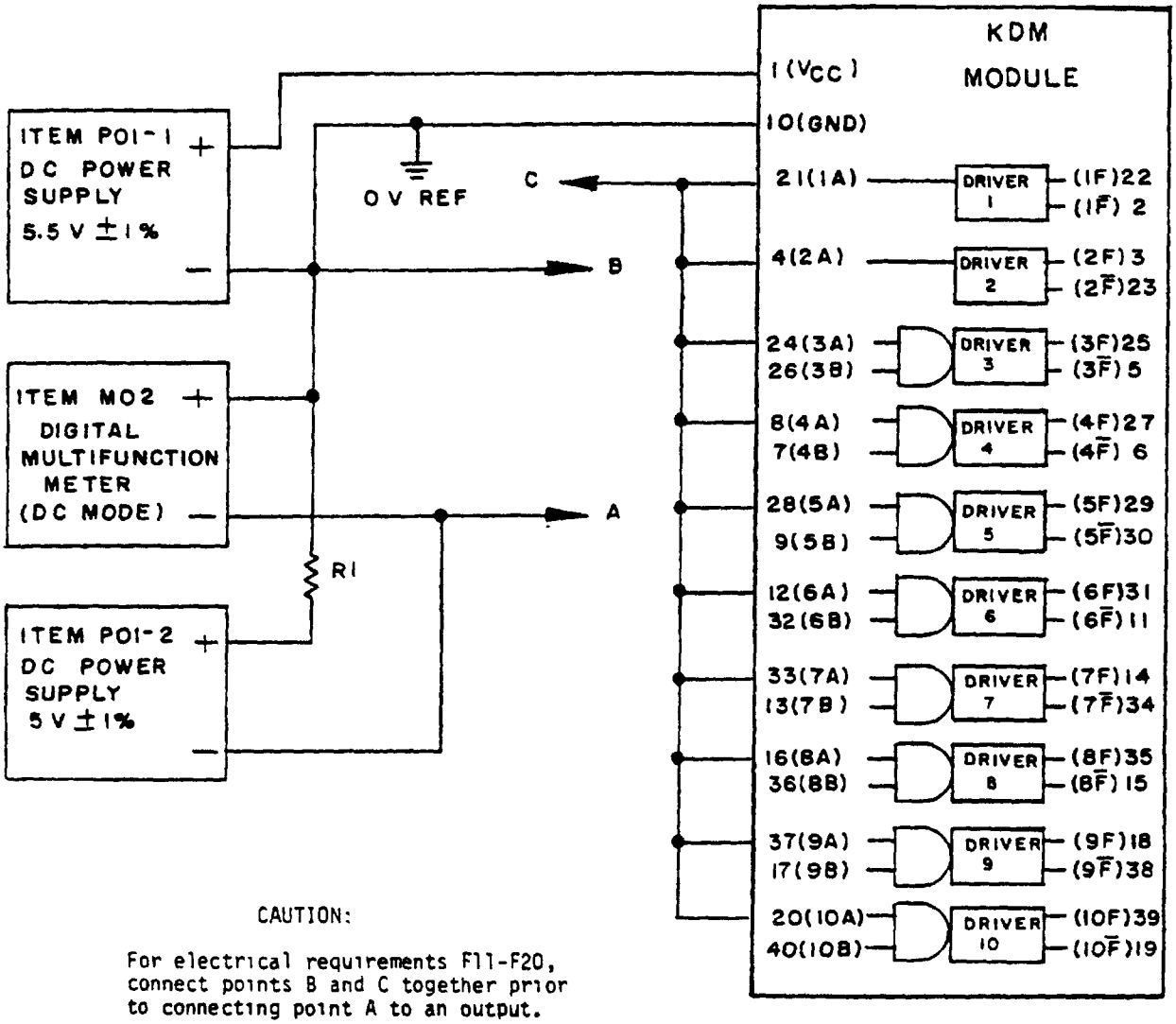
* FIGURE 9. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Output clamp diode voltage test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	$1\overline{F}$	Output clamp diode voltage	2	Open	Open	0.3	1.5	V	10
F2	$2\overline{F}$	"	23	"	"	"	"	"	"
F3	$3\overline{F}$	"	5	"	"	"	"	"	"
F4	$4\overline{F}$	"	6	"	"	"	"	"	"
F5	$5\overline{F}$	"	30	"	"	"	"	"	"
F6	$6\overline{F}$	"	11	"	"	"	"	"	"
F7	$7\overline{F}$	"	34	"	"	"	"	"	"
F8	$8\overline{F}$	"	15	"	"	"	"	"	"
F9	$9\overline{F}$	"	38	"	"	"	"	"	"
F10	$10\overline{F}$	"	19	"	"	"	"	"	"
F11	1F	"	22	C	B	"	"	"	"
F12	2F	"	3	"	"	"	"	"	"
F13	3F	"	25	"	"	"	"	"	"
F14	4F	"	27	"	"	"	"	"	"
F15	5F	"	29	"	"	"	"	"	"
F16	6F	"	31	"	"	"	"	"	"
F17	7F	"	14	"	"	"	"	"	"
F18	8F	"	35	"	"	"	"	"	"
F19	9F	"	18	"	"	"	"	"	"
F20	10F	"	39	"	"	"	"	"	"

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CAUTION:

For electrical requirements F11-F20, connect points B and C together prior to connecting point A to an output.

NOTE: R1 = 90.9Ω ±1%, 1/4 W.

* FIGURE 10. Output clamp diode voltage test circuit for table VIII.

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TABLE IX. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1F Output voltage, low level	Open	21	22	0	400	mV	11
G2	2F "	"	4	3	"	"	"	"
G3	3F "	"	24	25	"	"	"	"
G4	3F "	"	26	25	"	"	"	"
G5	4F "	"	8	27	"	"	"	"
G6	4F "	"	7	27	"	"	"	"
G7	5F "	"	28	29	"	"	"	"
G8	5F "	"	9	29	"	"	"	"
G9	6F "	"	12	31	"	"	"	"
G10	6F "	"	32	31	"	"	"	"
G11	7F "	"	33	14	"	"	"	"
G12	7F "	"	13	14	"	"	"	"
G13	8F "	"	16	35	"	"	"	"
G14	8F "	"	36	35	"	"	"	"
G15	9F "	"	37	18	"	"	"	"
G16	9F "	"	17	18	"	"	"	"
G17	10F "	"	20	39	"	"	"	"
G18	10F "	"	40	39	"	"	"	"
G19	$1\bar{F}$ "	21	Open	2	"	"	"	"
G20	$2\bar{F}$ "	4	Open	23	"	"	"	"

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TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G21	$3\overline{F}$	Output voltage, low level	24, 26	Open	5	0	400	mV	11
G22	$4\overline{F}$	"	8, 7	"	6	"	"	"	"
G23	$5\overline{F}$	"	28, 9	"	30	"	"	"	"
G24	$6\overline{F}$	"	12, 32	"	11	"	"	"	"
G25	$7\overline{F}$	"	33, 13	"	34	"	"	"	"
G26	$8\overline{F}$	"	16, 36	"	15	"	"	"	"
G27	$9\overline{F}$	"	37, 17	"	38	"	"	"	"
G28	$10\overline{F}$	"	20, 40	"	19	"	"	"	"

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TABLE X. Output voltage (high-level) test.

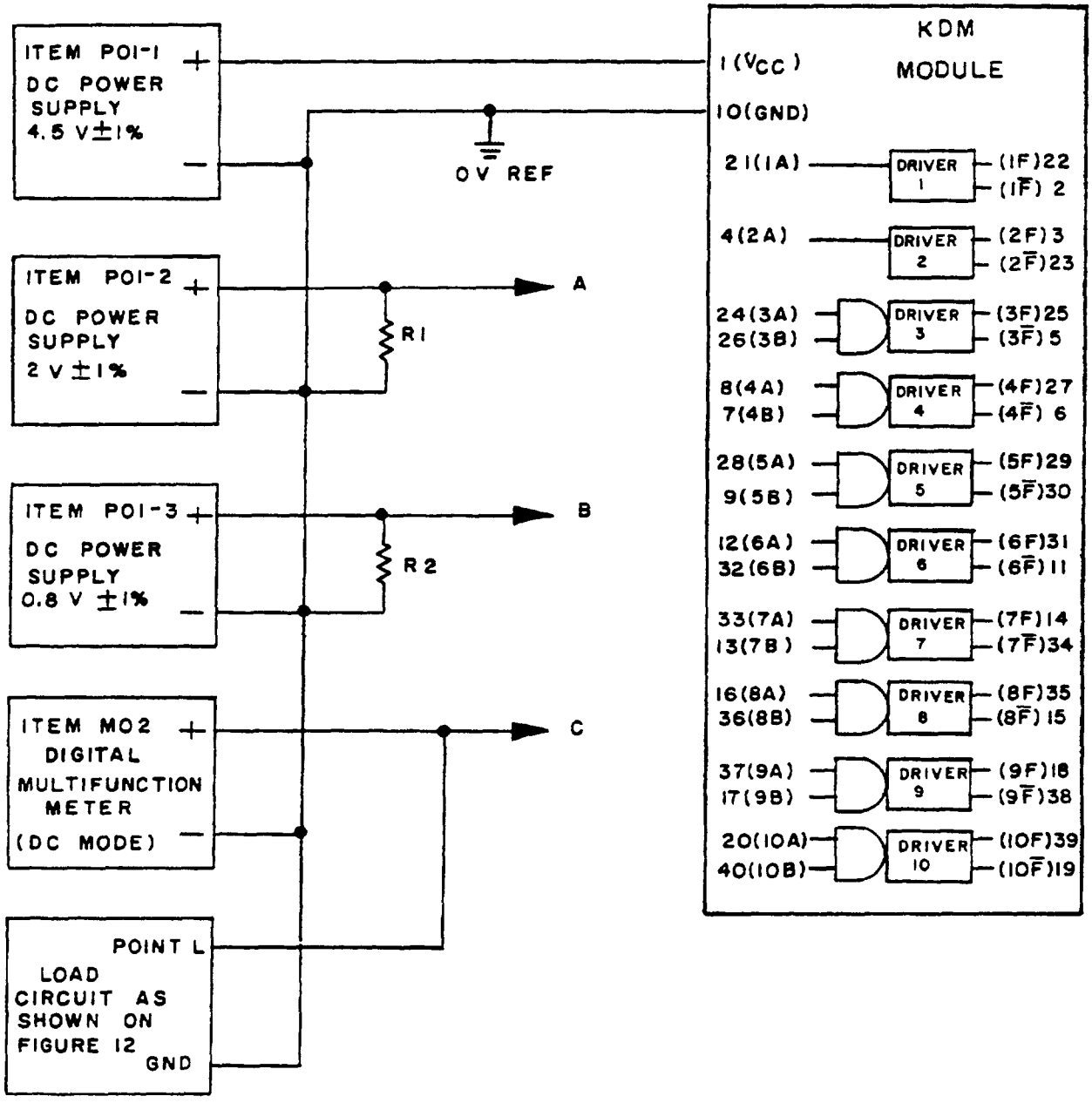
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1F	Output voltage, high level	21	Open	22	2.4	4.5	V	11
H2	2F	"	4	"	3	"	"	"	"
H3	3F	"	24, 26	"	25	"	"	"	"
H4	4F	"	8, 7	"	27	"	"	"	"
H5	5F	"	28, 9	"	29	"	"	"	"
H6	6F	"	12, 32	"	31	"	"	"	"
H7	7F	"	33, 13	"	14	"	"	"	"
H8	8F	"	16, 36	"	35	"	"	"	"
H9	9F	"	37, 17	"	18	"	"	"	"
H10	10F	"	20, 40	"	39	"	"	"	"
H11	$1\overline{F}$	"	Open	21	2	"	"	"	"
H12	$2\overline{F}$	"	"	4	23	"	"	"	"
H13	$3\overline{F}$	"	"	24	5	"	"	"	"
H14	$3\overline{F}$	"	"	26	5	"	"	"	"
H15	$4\overline{F}$	"	"	8	6	"	"	"	"
H16	$4\overline{F}$	"	"	7	6	"	"	"	"
H17	$5\overline{F}$	"	"	28	30	"	"	"	"
H18	$5\overline{F}$	"	"	9	30	"	"	"	"
H19	$6\overline{F}$	"	"	12	11	"	"	"	"
H20	$6\overline{F}$	"	"	32	11	"	"	"	"

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TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H21	$7\overline{F}$	Output voltage, high level	Open	33	34	2.4	4.5	V	11
H22	$7\overline{F}$	"	"	13	34	"	"	"	"
H23	$8\overline{F}$	"	"	16	15	"	"	"	"
H24	$8\overline{F}$	"	"	36	15	"	"	"	"
H25	$9\overline{F}$	"	"	37	38	"	"	"	"
H26	$9\overline{F}$	"	"	17	38	"	"	"	"
H27	$10\overline{F}$	"	"	20	19	"	"	"	"
H28	$10\overline{F}$	"	"	40	19	"	"	"	"

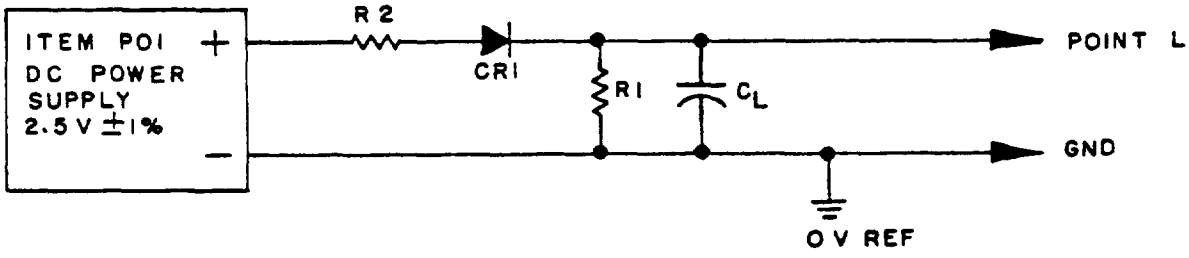
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. Load circuit 1 (dc) for test conditions G and H:
   a. CR1 = 1N4150, or equivalent.
   b. R1 = 243Ω ±1%, 1/4 W.
   c. R2 = Select value for a current flow of 40 mA ±1% with point L held at 0.4 V.
   d. C_L is not required.
2. Load circuit 2 (ac) for test conditions N, P, R, S, T, and U:
   a. CR1 = 1N4150, or equivalent.
   b. R1 = 957Ω ±1%, 1/4 W.
   c. R2 = Select value for a current flow of 40 mA ±1% with point L held at 0.4 V.
   d. For test conditions N, P, R, and S, C_L = 100 ±5 pF (including probe and parasitic capacitance).
   e. For test conditions T and U, C_L = 50 ±5 pF (including probe and parasitic capacitance).
   f. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
   g. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV peak.

FIGURE 12. KDM logic load circuit.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25 C ±5 C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1A Input current, low level	Open	21	Open	0.40	1.73	mA	13
J2	2A "	"	4	"	"	"	"	"
J3	3A "	"	24	"	"	"	"	"
J4	3B "	"	26	"	"	"	"	"
J5	4A "	"	8	"	"	"	"	"
J6	4B "	"	7	"	"	"	"	"
J7	5A "	"	28	"	"	"	"	"
J8	5B "	"	9	"	"	"	"	"
J9	6A "	"	12	"	"	"	"	"
J10	6B "	"	32	"	"	"	"	"
J11	7A "	"	33	"	"	"	"	"
J12	7B "	"	13	"	"	"	"	"
J13	8A "	"	16	"	"	"	"	"
J14	8B "	"	36	"	"	"	"	"
J15	9A "	"	37	"	"	"	"	"
J16	9B "	"	17	"	"	"	"	"
J17	10A "	"	20	"	"	"	"	"
J18	10B "	"	40	"	"	"	"	"

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TABLE XII. Input current (high-level) test.

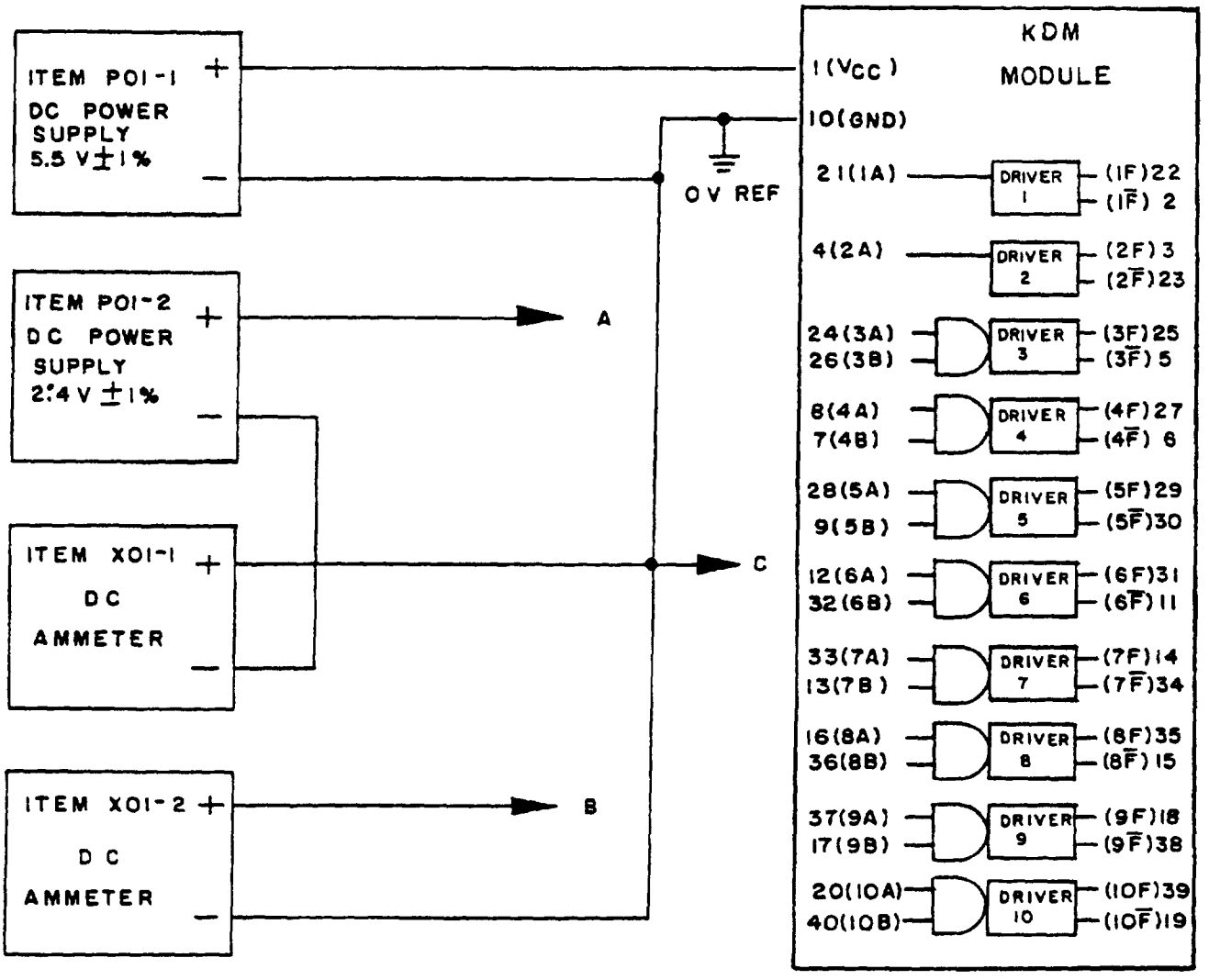
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	1A	Input current, high level	21	Open	Open	0	90	µA	13
K2	2A	"	4	"	"	"	90	"	"
K3	3A	"	24	"	"	"	30	"	"
K4	4A	"	8	"	"	"	"	"	"
K5	5A	"	28	"	"	"	"	"	"
K6	6A	"	12	"	"	"	"	"	"
K7	7A	"	33	"	"	"	"	"	"
K8	8A	"	16	"	"	"	"	"	"
K9	9A	"	37	"	"	"	"	"	"
K10	10A	"	20	"	"	"	"	"	"
K11	3B	"	26	"	"	"	60	"	"
K12	4B	"	7	"	"	"	"	"	"
K13	5B	"	9	"	"	"	"	"	"
K14	6B	"	32	"	"	"	"	"	"
K15	7B	"	13	"	"	"	"	"	"
K16	8B	"	36	"	"	"	"	"	"
K17	9B	"	17	"	"	"	"	"	"
K18	10B	"	40	"	"	"	"	"	"

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TABLE XIII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1F Output short circuit current	Open	22	Open	40	120	mA	13
L2	2F "	"	3	"	"	"	"	"
L3	3F "	"	25	"	"	"	"	"
L4	4F "	"	27	"	"	"	"	"
L5	5F "	"	29	"	"	"	"	"
L6	6F "	"	31	"	"	"	"	"
L7	7F "	"	14	"	"	"	"	"
L8	8F "	"	35	"	"	"	"	"
L9	9F "	"	18	"	"	"	"	"
L10	10F "	"	39	"	"	"	"	"
L11	$1\overline{F}$ "	"	2	21	"	"	"	"
L12	$2\overline{F}$ "	"	23	4	"	"	"	"
L13	$3\overline{F}$ "	"	5	24	"	"	"	"
L14	$4\overline{F}$ "	"	6	8	"	"	"	"
L15	$5\overline{F}$ "	"	30	28	"	"	"	"
L16	$6\overline{F}$ "	"	11	12	"	"	"	"
L17	$7\overline{F}$ "	"	34	33	"	"	"	"
L18	$8\overline{F}$ "	"	15	16	"	"	"	"
L19	$9\overline{F}$ "	"	38	37	"	"	"	"
L20	$10\overline{F}$ "	"	19	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. Use item X01-2 for tables XI and XIII; use item X01-1 for table XII.
3. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken with 5 seconds.

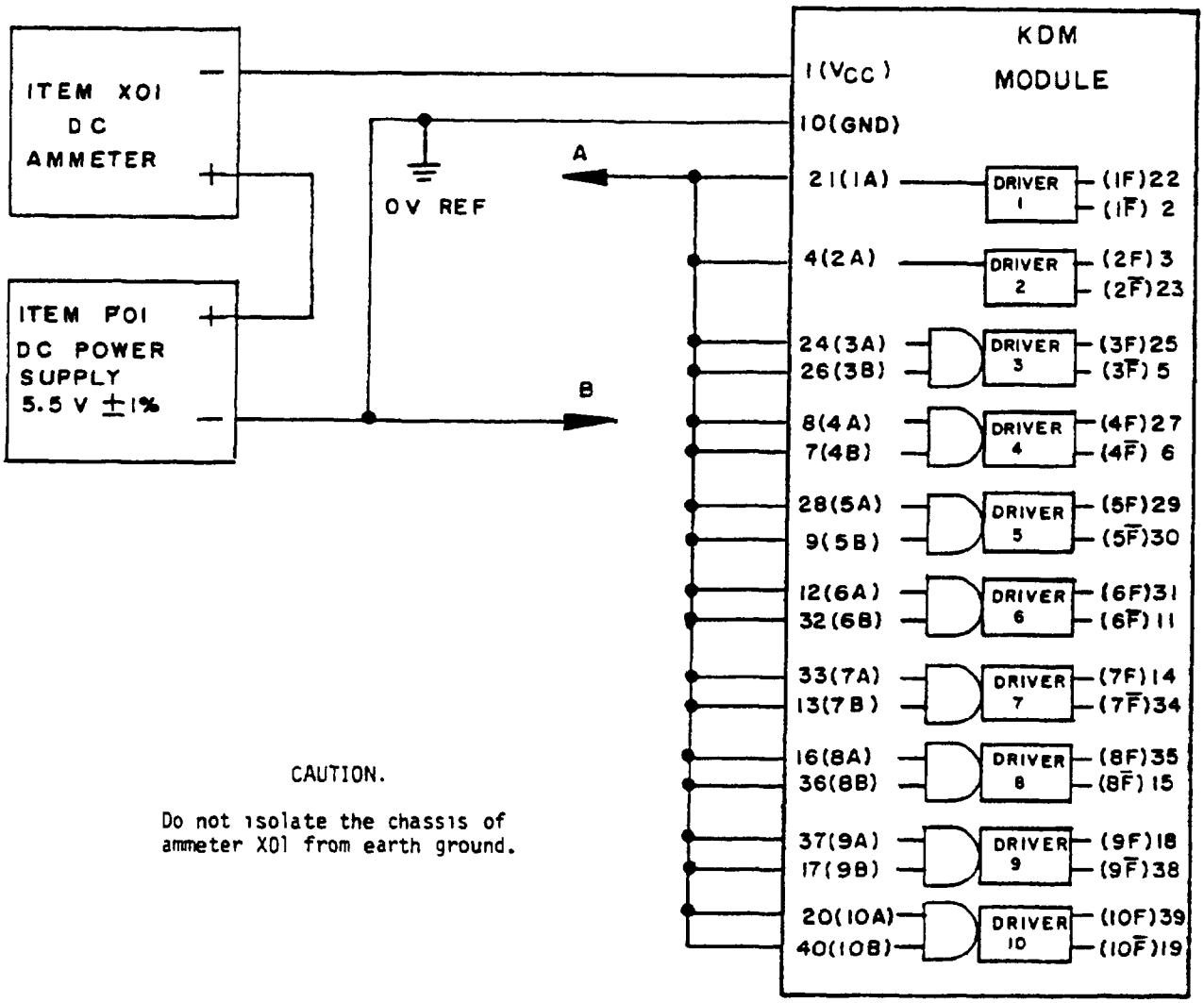
* FIGURE 13. Input current (low-level and high-level) and output short circuit current test circuit for tables XI, XII, and XIII.

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TABLE XIV. Power supply current (low-level and high-level) tests.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	Open	Open	1	244	mA	14
M2	Power supply current, high level	B	A	1	244	mA	14

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CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 14. Power supply current (low-level and high-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N1	1F	Transition time (t_{TLH})	21	Open	22	4	45	ns	15,16
N2	$1\overline{F}$	"	21	"	2	"	"	"	"
N3	2F	"	4	"	3	"	"	"	"
N4	$2\overline{F}$	"	4	"	23	"	"	"	"
N5	3F	"	26	24	25	"	"	"	"
N6	$3\overline{F}$	"	26	24	5	"	"	"	"
N7	4F	"	7	8	27	"	"	"	"
N8	$4\overline{F}$	"	7	8	6	"	"	"	"
N9	5F	"	9	28	29	"	"	"	"
N10	$5\overline{F}$	"	9	28	30	"	"	"	"
N11	6F	"	32	12	31	"	"	"	"
N12	$6\overline{F}$	"	32	12	11	"	"	"	"
N13	7F	"	13	33	14	"	"	"	"
N14	$7\overline{F}$	"	13	33	34	"	"	"	"
N15	8F	"	36	16	35	"	"	"	"
N16	$8\overline{F}$	"	36	16	15	"	"	"	"
N17	9F	"	17	37	18	"	"	"	"
N18	$9\overline{F}$	"	17	37	38	"	"	"	"
N19	10F	"	40	20	39	"	"	"	"
N20	$10\overline{F}$	"	40	20	19	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	1F	Transition time (t_{THL})	21	Open	22	2	20	ns	15,16
P2	$1\overline{F}$	"	21	"	2	"	"	"	"
P3	2F	"	4	"	3	"	"	"	"
P4	$2\overline{F}$	"	4	"	23	"	"	"	"
P5	3F	"	26	24	25	"	"	"	"
P6	$3\overline{F}$	"	26	24	5	"	"	"	"
P7	4F	"	7	8	27	"	"	"	"
P8	$4\overline{F}$	"	7	8	6	"	"	"	"
P9	5F	"	9	28	29	"	"	"	"
P10	$5\overline{F}$	"	9	28	30	"	"	"	"
P11	6F	"	32	12	31	"	"	"	"
P12	$6\overline{F}$	"	32	12	11	"	"	"	"
P13	7F	"	13	33	14	"	"	"	"
P14	$7\overline{F}$	"	13	33	34	"	"	"	"
P15	8F	"	36	16	35	"	"	"	"
P16	$8\overline{F}$	"	36	16	15	"	"	"	"
P17	9F	"	17	37	18	"	"	"	"
P18	$9\overline{F}$	"	17	37	38	"	"	"	"
P19	10F	"	40	20	39	"	"	"	"
P20	$10\overline{F}$	"	40	20	19	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

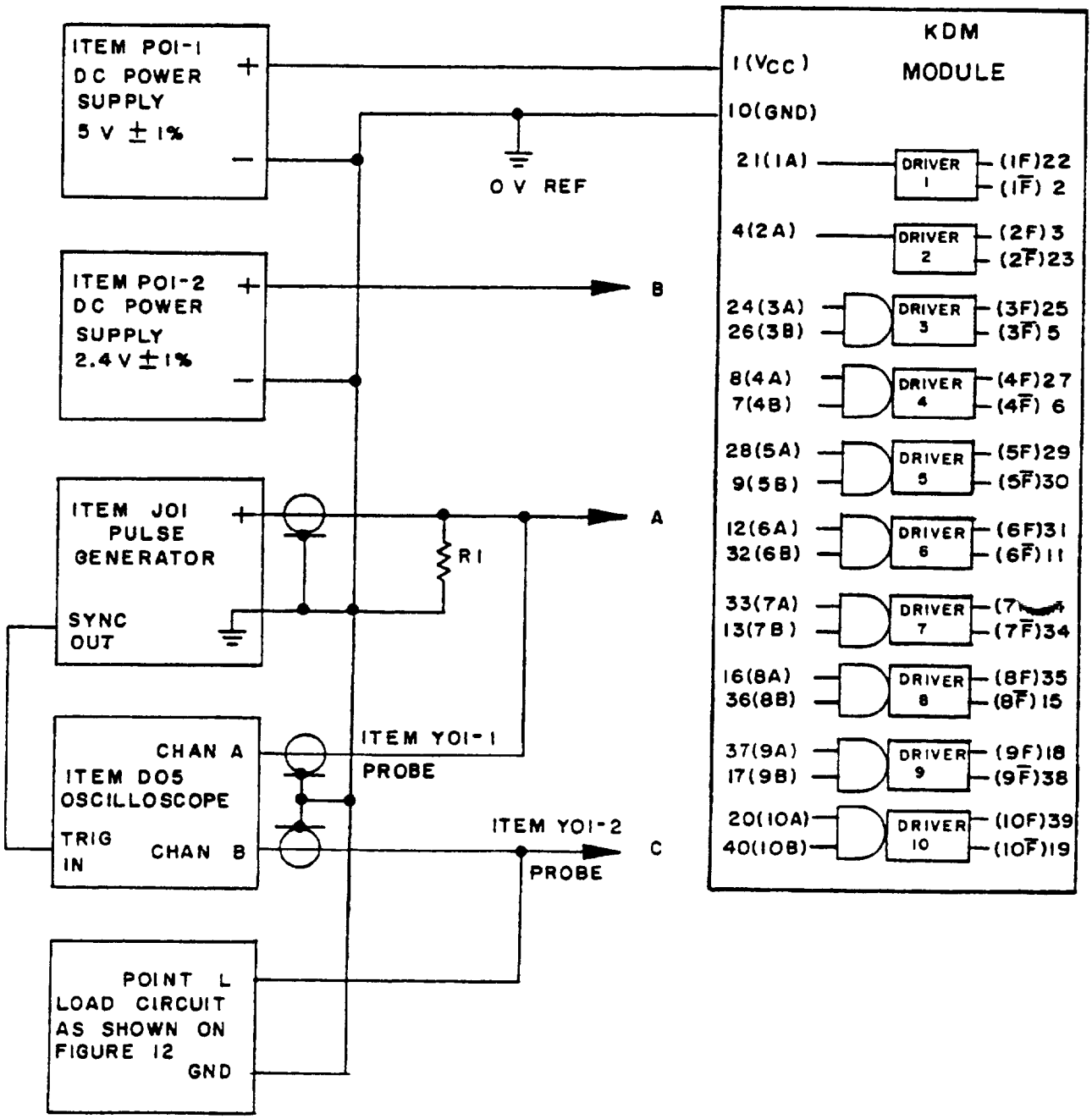
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	1F	Propagation delay time (t_{PLH})	21	Open	22	5	30	ns	15,16
R2	1$\overline{F}$	"	21	"	2	"	"	"	"
R3	2F	"	4	"	3	"	"	"	"
R4	2$\overline{F}$	"	4	"	23	"	"	"	"
R5	3F	"	26	24	25	"	"	"	"
R6	3$\overline{F}$	"	26	24	5	"	"	"	"
R7	4F	"	7	8	27	"	"	"	"
R8	4$\overline{F}$	"	7	8	6	"	"	"	"
R9	5F	"	9	28	29	"	"	"	"
R10	5$\overline{F}$	"	9	28	30	"	"	"	"
R11	6F	"	32	12	31	"	"	"	"
R12	6$\overline{F}$	"	32	12	11	"	"	"	"
R13	7F	"	13	33	14	"	"	"	"
R14	7$\overline{F}$	"	13	33	34	"	"	"	"
R15	8F	"	36	16	35	"	"	"	"
R16	8$\overline{F}$	"	36	16	15	"	"	"	"
R17	9F	"	17	37	18	"	"	"	"
R18	9$\overline{F}$	"	17	37	38	"	"	"	"
R19	10F	"	40	20	39	"	"	"	"
R20	10$\overline{F}$	"	40	20	19	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S1	1F	Propagation delay time (t_{PHL})	21	Open	22	5	30	ns	15,16
S2	$1\overline{F}$	"	21	"	2	"	"	"	"
S3	2F	"	4	"	3	"	"	"	"
S4	$2\overline{F}$	"	4	"	23	"	"	"	"
S5	3F	"	26	24	25	"	"	"	"
S6	$3\overline{F}$	"	26	24	5	"	"	"	"
S7	4F	"	7	8	27	"	"	"	"
S8	$4\overline{F}$	"	7	8	6	"	"	"	"
S9	5F	"	9	28	29	"	"	"	"
S10	$5\overline{F}$	"	9	28	30	"	"	"	"
S11	6F	"	32	12	31	"	"	"	"
S12	$6\overline{F}$	"	32	12	11	"	"	"	"
S13	7F	"	13	33	14	"	"	"	"
S14	$7\overline{F}$	"	13	33	34	"	"	"	"
S15	8F	"	36	16	35	"	"	"	"
S16	$8\overline{F}$	"	36	16	15	"	"	"	"
S17	9F	"	17	37	18	"	"	"	"
S18	$9\overline{F}$	"	17	37	38	"	"	"	"
S19	10F	"	40	20	39	"	"	"	"
S20	$10\overline{F}$	"	40	20	19	"	"	"	"

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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

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TABLE XVI. Output crosstalk (low-level) test - Continued.

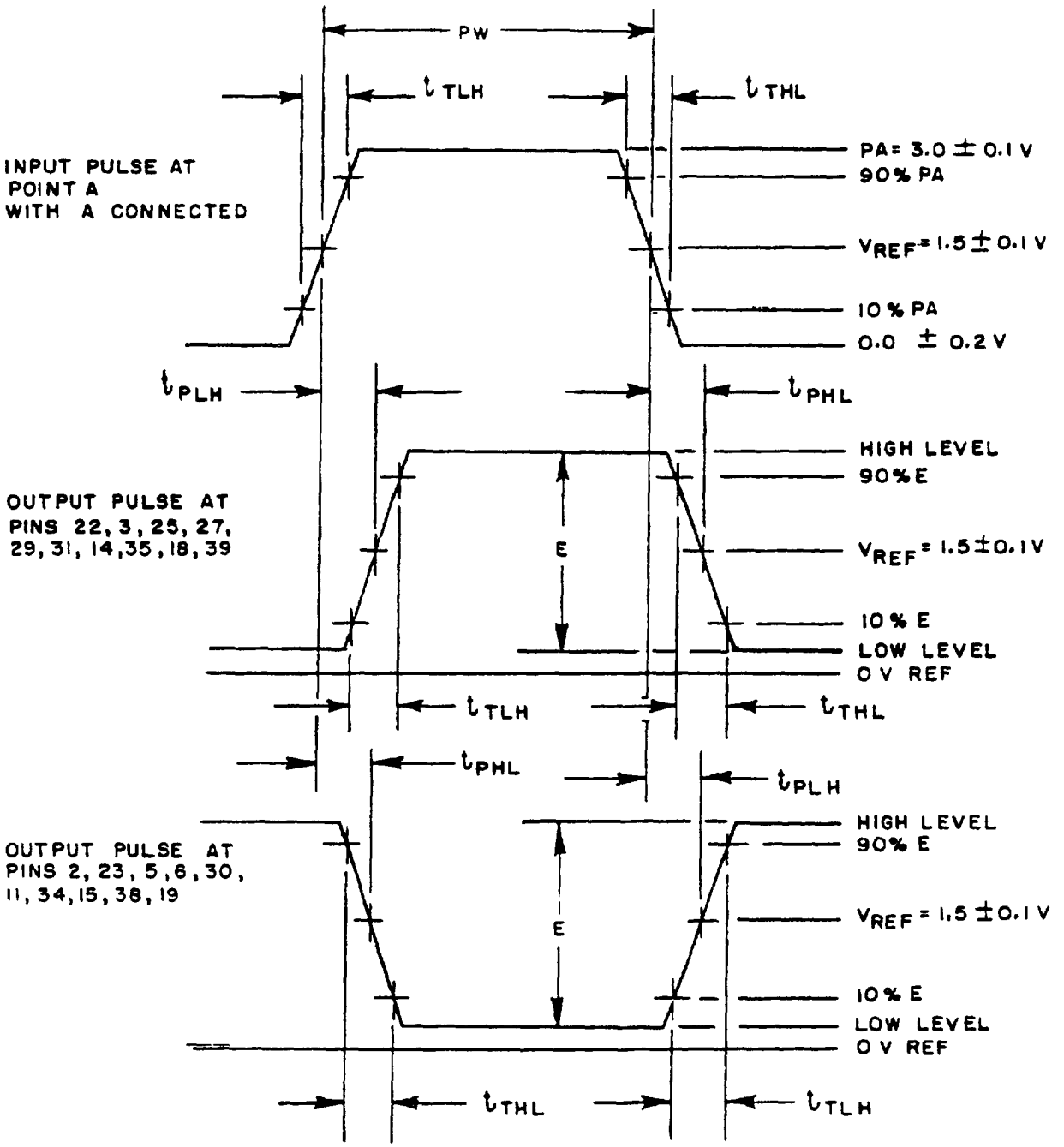
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T10	5F	Output crosstalk, low level	4,21,24,26,8,7,12, 32,33,13,16,36,37, 17,20,40	28,9	Open	30	0	1	V	17,18
T11	6F	"	4,21,24,26,8,7,28, 9,33,13,16,36,37, 17,20,40	Open	12,32	31	"	"	"	"
T12	6F	"	4,21,24,26,8,7,28, 9,33,13,16,36,37, 17,20,40	12,32	Open	11	"	"	"	"
T13	7F	"	4,21,24,26,8,7,28, 9,12,32,16,36,37, 17,20,40	Open	33,13	14	"	"	"	"
T14	7F	"	4,21,24,26,8,7,28, 9,12,32,16,36,37, 17,20,40	33,13	Open	34	"	"	"	"
T15	8F	"	4,21,24,26,8,7,28, 9,12,32,33,13,37, 17,20,40	Open	16,36	35	"	"	"	"
T16	8F	"	4,21,24,26,8,7,28, 9,12,32,33,13,37, 17,20,40	16,36	Open	15	"	"	"	"
T17	9F	"	4,21,24,26,8,7,28, 9,12,32,33,13,16,36, 20,40	Open	37,17	18	"	"	"	"
T18	9F	"	4,21,24,26,8,7,28, 9,12,32,33,13,16,36, 20,40	32,17	Open	38	"	"	"	"

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TABLE XVI. Output crosstalk (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T19	10F	Output crosstalk, low level	4,21,24,26,8,7,28, 9,12,32,33,13,16, 36,37,17	Open	20,40	39	0	1	V	17,18
T20	$10\overline{F}$	Output crosstalk, low level	4,21,24,26,8,7,28, 9,12,32,33,13,16, 36,37,17	20,40	Open	19	0	1	V	17,18

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NOTES:
1. Input pulse parameters:
   PW = 110 ±10 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 500 kHz ±10%.
2. Before making measurements, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XV.

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TABLE XVI. Output crosstalk (low-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1F	Output crosstalk, low level	4,24,26,8,7,28,9, 12,32,33,13,16, 36,37,17,20,40	Open	21	22	0	1	V	17,18
T2	$1\bar{F}$	"	4,24,26,8,7,28,9, 12,32,33,13,16, 36,37,17,20,40	21	Open	2	"	"	"	"
T3	2F	"	21,24,26,8,7,28, 9,12,32,33,13,16, 36,37,17,20,40	Open	4	3	"	"	"	"
T4	$2\bar{F}$	"	21,24,26,8,7,28, 9,12,32,33,13,16, 36,37,17,20,40	4	Open	23	"	"	"	"
T5	3F	"	4,21,8,7,28,9,12, 32,33,13,16,36, 37,17,20,40	Open	24,26	25	"	"	"	"
T6	$3\bar{F}$	"	4,21,8,7,28,9,12, 32,33,13,16,36, 37,17,20,40	24,26	Open	5	"	"	"	"
T7	4F	"	4,21,24,26,28,9, 12,32,33,13,16, 36,37,17,20,40	Open	8,7	27	"	"	"	"
T8	$4\bar{F}$	"	4,21,24,26,28,9, 12,32,33,13,16, 36,37,17,20,40	8,7	Open	6	"	"	"	"
T9	5F	"	4,21,24,26,8,7, 12,32,33,13,16, 36,37,17,20,40	Open	28,9	29	"	"	"	"

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1F	Output crosstalk, high level	4,24,26,8,7,28,9, 12,32,33,13,16, 36,37,17,20,40	21	Open	22	0	750	mV	17,18
U2	1$\overline{F}$	"	4,24,26,8,7,28,9, 12,32,33,13,16, 36,37,17,20,40	Open	21	2	"	"	"	"
U3	2F	"	21,24,26,8,7,28,9, 12,32,33,13,16,36, 37,17,20,40	4	Open	3	"	"	"	"
U4	2$\overline{F}$	"	21,24,26,8,7,28,9, 12,32,33,13,16,36, 37,17,20,40	Open	4	23	"	"	"	"
U5	3F	"	4,21,8,7,28,9,12, 32,33,13,16,36, 37,17,20,40	24,26	Open	25	"	"	"	"
U6	3$\overline{F}$	"	4,21,8,7,28,9,12, 32,33,13,16,36, 37,17,20,40	Open	24,26	5	"	"	"	"
U7	4F	"	4,21,24,26,28,9,12, 32,33,13,16,36,37, 17,20,40	8,7	Open	27	"	"	"	"
U8	4$\overline{F}$	"	4,21,24,26,28,9,12, 32,33,13,16,36,37, 17,20,40	Open	8,7	6	"	"	"	"
U9	5F	"	4,21,24,26,8,7,12, 32,33,13,16,36,37, 17,20,40	28,9	Open	29	"	"	"	"

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TABLE XVII. Output crosstalk (high-level) test - Continued.

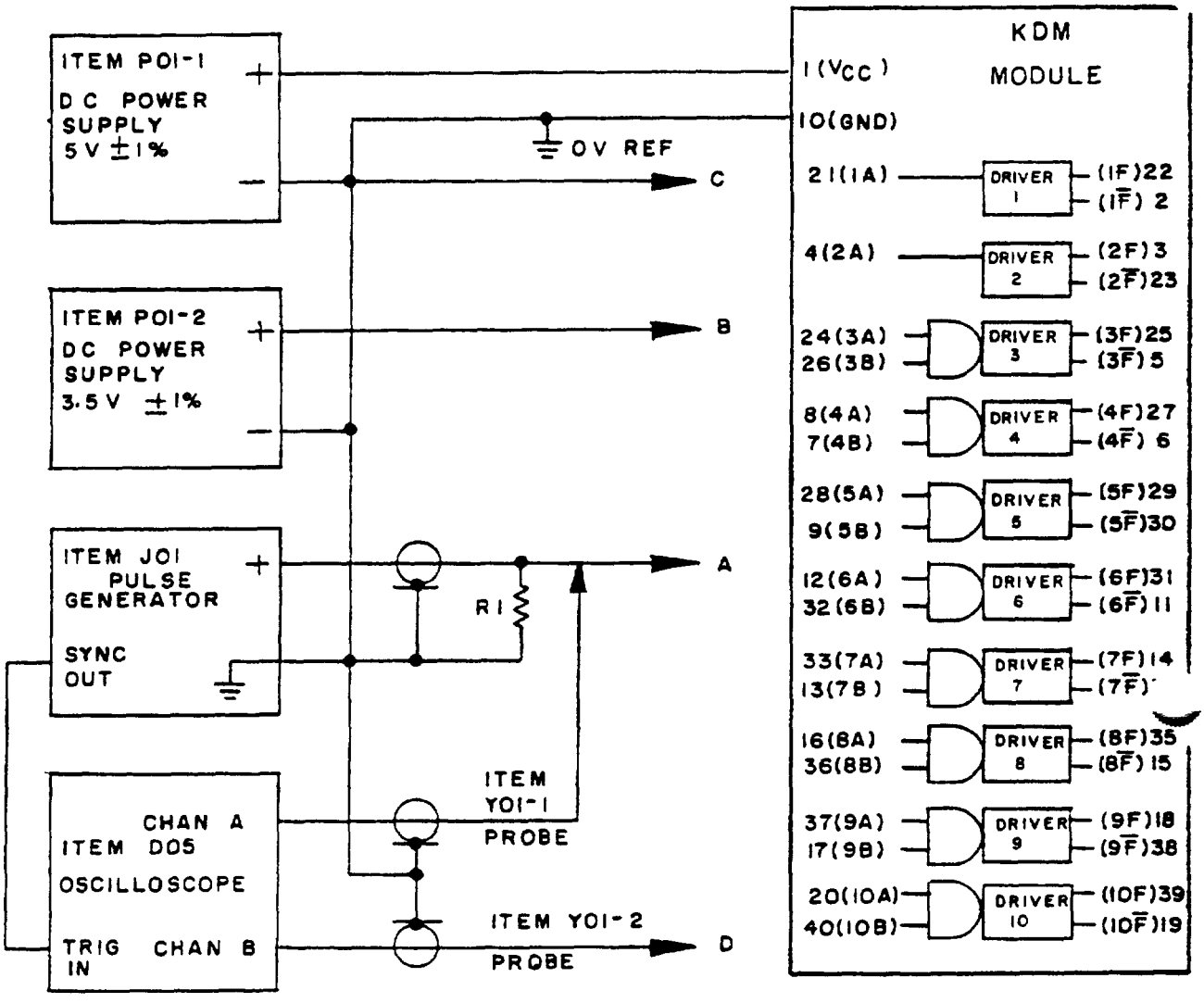
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U10	5F̄	Output crosstalk, high level	4,21,24,26,8,7,12, 32,33,13,16,36,37, 17,20,40	Open	28,9	30	0	750	mV	17,18
U11	6F	"	4,21,24,26,8,7,28, 9,33,13,16,36,37, 17,20,40	12,32	Open	31	"	"	"	"
U12	6F̄	"	4,21,24,26,8,7,28, 9,33,13,16,36,37, 17,20,40	Open	12,32	11	"	"	"	"
U13	7F	"	4,21,24,26,8,7,28, 9,12,32,16,36,37, 17,20,40	33,13	Open	14	"	"	"	"
U14	7F̄	"	4,21,24,26,8,7,28, 9,12,32,16,36,37, 17,20,40	Open	33,13	34	"	"	"	"
U15	8F	"	4,21,24,26,8,7,28, 9,12,32,33,13,37, 17,20,40	16,36	Open	35	"	"	"	"
U16	8F̄	"	4,21,24,26,8,7,28, 9,12,32,33,13,37, 17,20,40	Open	16,36	15	"	"	"	"
U17	9F	"	4,21,24,26,8,7,28, 9,12,32,33,13,16, 36,20,40	37,17	Open	18	"	"	"	"
U18	9F̄	"	4,21,24,26,8,7,28, 9,12,32,33,13,16, 36,20,40	Open	37,17	38	"	"	"	"

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TABLE XVII. Output crosstalk (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U19	10F	Output crosstalk, high level	4,21,24,26,8,7,28, 9,12,32,33,13,16, 36,37,17	20,40	Open	39	0	750	mV	17,18
U20	$10\bar{F}$	Output crosstalk, high level	4,21,24,26,8,7,28, 9,12,32,33,13,16, 36,37,17	Open	20,40	19	0	750	mV	17,18

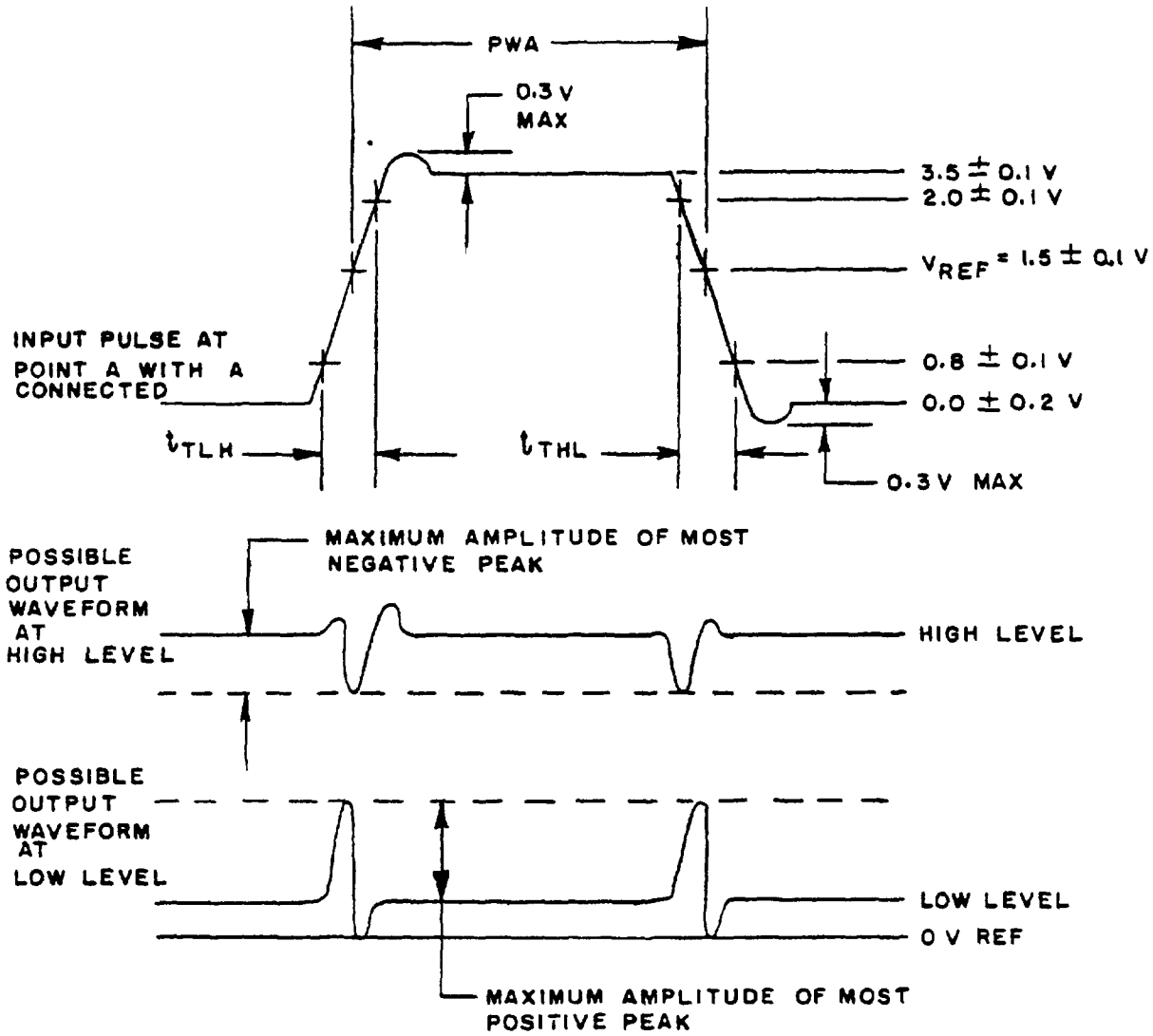
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NOTES.
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 12.

FIGURE 17. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTE: Input pulse parameters:
PWA = 200 ±20 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 MHz ±10%.

FIGURE 18. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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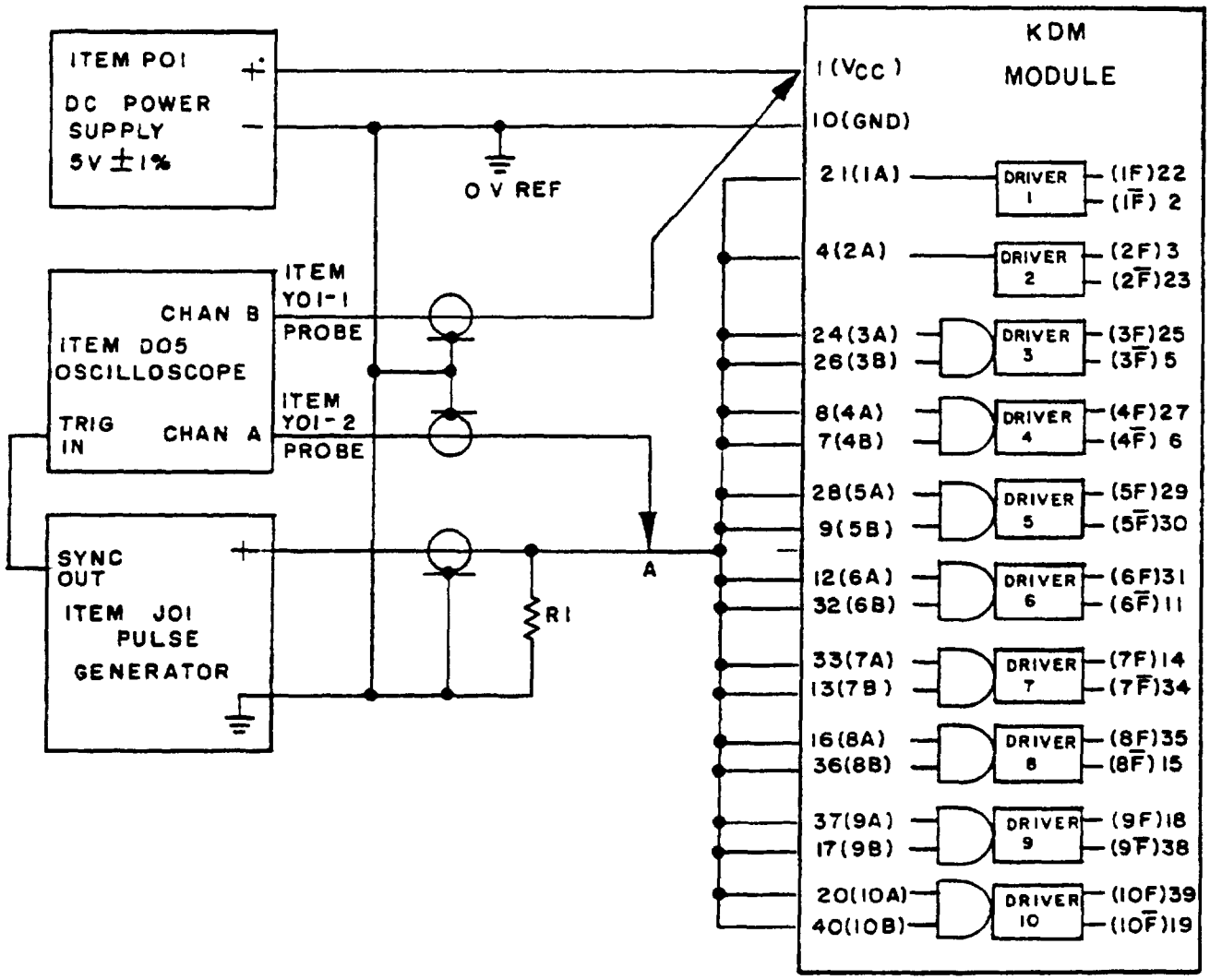
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0.0	1.3	V	19,20

TABLE XIX. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
W	Power supply transient rise time	4.8	8.0	ms	21,22

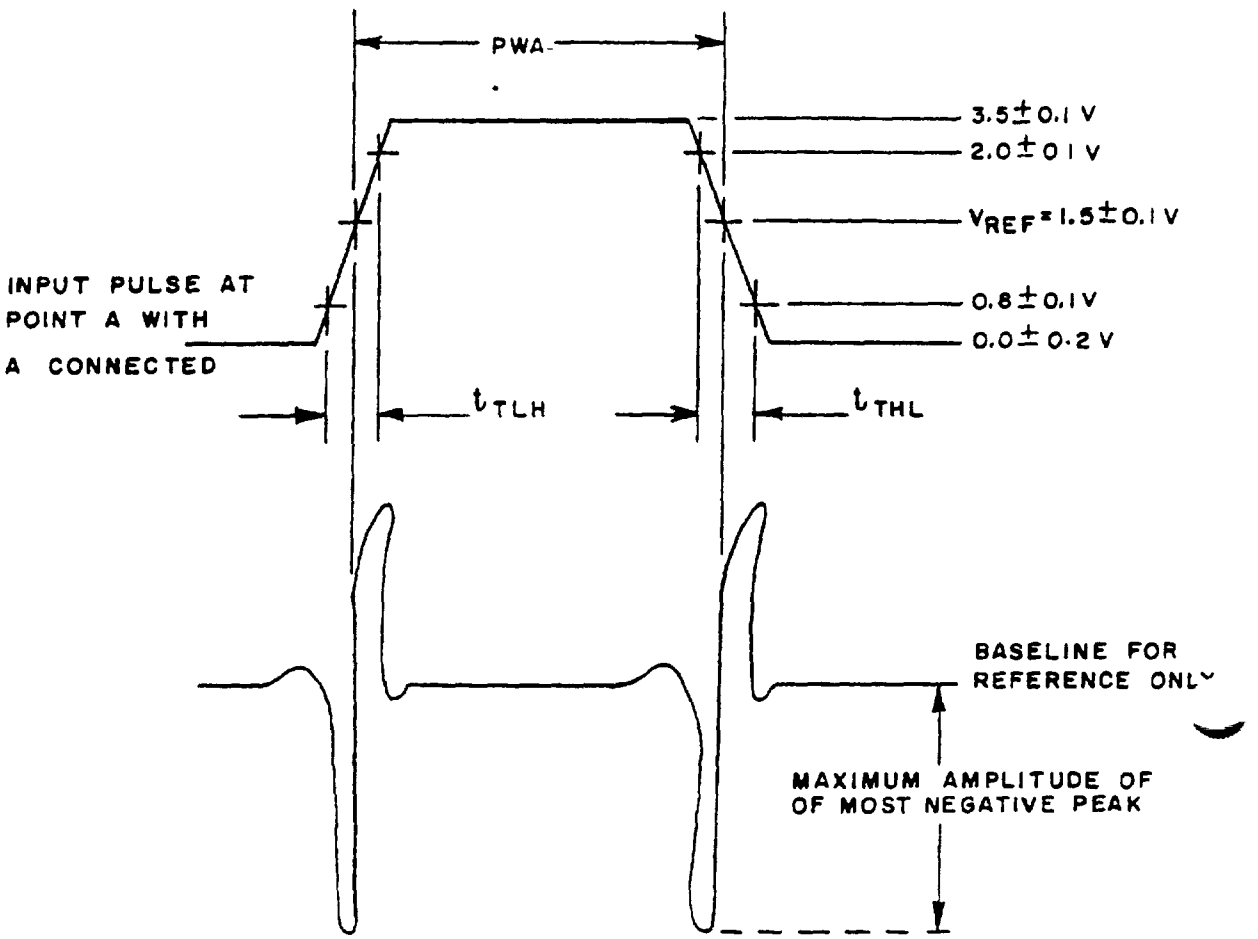
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 19. Power supply transient amplitude test circuit for table XVIII.

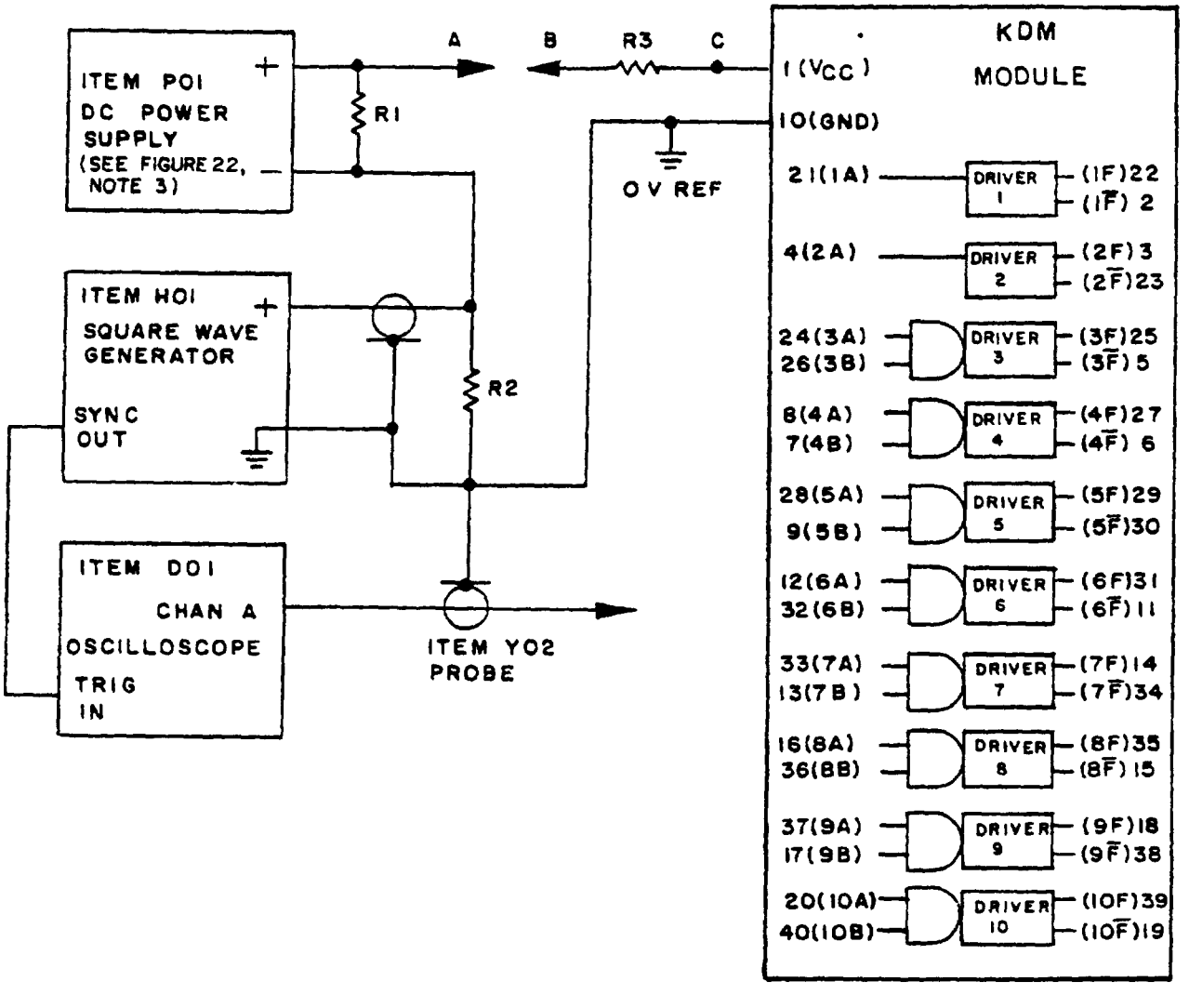
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NOTE: Input pulse parameters:
PWA = 200 ±20 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1MHz ±10%.

FIGURE 20. Power supply transient amplitude test circuit waveforms for table XVIII.

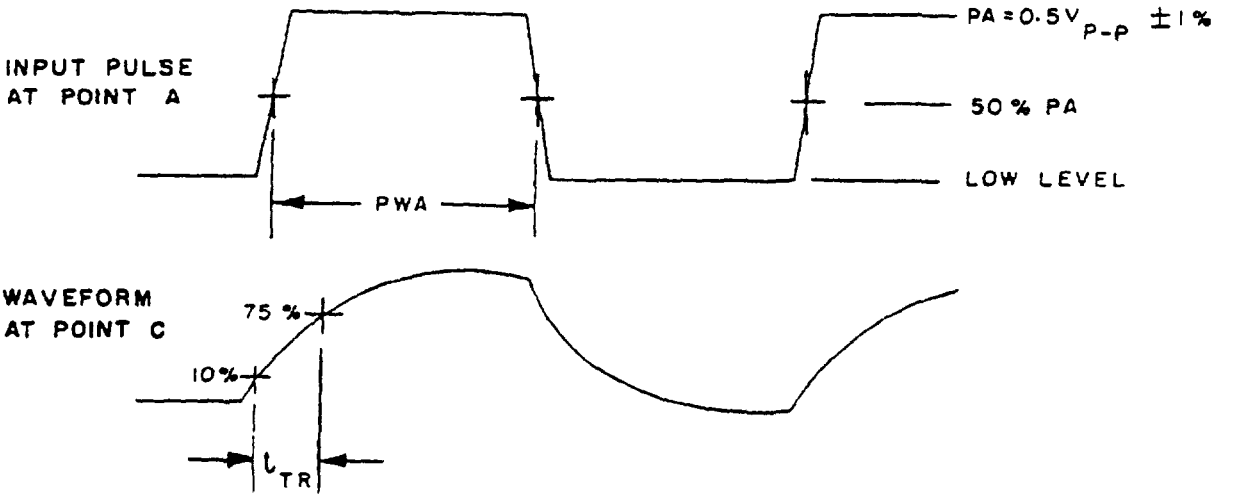
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 21. Power supply transient rise time test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   PWA = 40 ms ±10%.
   PRR = 12.5 Hz (reference).
   $t_{THL} = t_{TLH} \leq 100$ ns (reference).
2. With point A open, adjust input as shown.
3. With point A open, adjust item P01 such that the low level of the input is at 0.00 ±0.01 V.
4. Connect A to B, recheck the low level and pulse parameters at point A.
5. Measure transient rise time (t_{TR}) at point C.

FIGURE 22. Power supply transient rise time test circuit waveforms for table XIX.

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